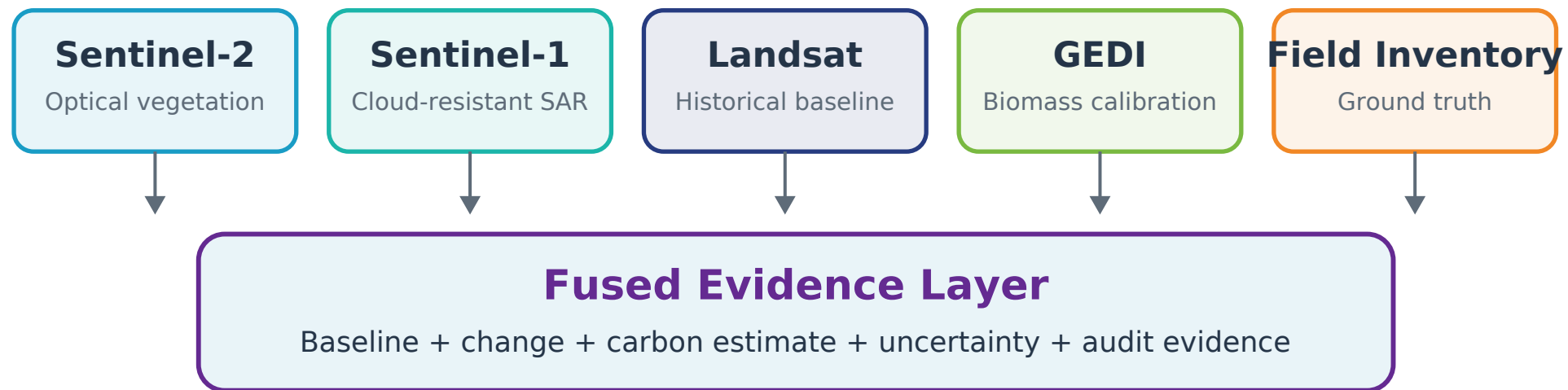




Best Free Core Dataset Stack

AI-Powered Carbon Additionality, Plantation Monitoring and Forest Change Detection
Candidate Interview Revision Guide



Core principle: combine open satellite data with project records and field measurements. No single dataset can prove additionality or certification-grade carbon on its own.

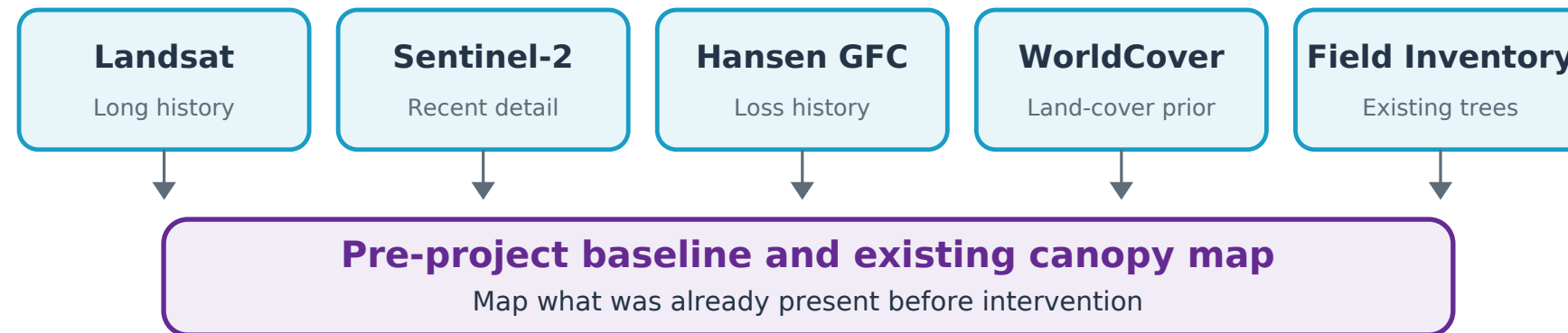
1. Best Free Core Stack

Use this combination as the default MVP architecture. Each dataset answers a different part of the carbon-monitoring problem.

Dataset	Primary role	Why it is essential
Sentinel-2 Surface Reflectance	Optical vegetation monitoring	10 m visible/NIR bands, red-edge and SWIR information, approximately 5-day revisit; suitable for canopy, plantation and vegetation-index analysis.
Sentinel-1 GRD	All-weather disturbance monitoring	Radar supports monitoring through cloud and haze; useful for sudden canopy change and temporal continuity.
Landsat Collection 2 Level-2	Long historical baseline	30 m surface reflectance and decades of land history help test whether vegetation or degradation existed before project registration.
GEDI lidar products	Biomass and canopy calibration	Footprint observations of forest height, structure and biomass strengthen models that convert spectral signals into carbon estimates.
Hansen Global Forest Change	Historical forest-loss screening	Annual loss history supports disturbance risk, baseline checks and reference-area matching.
Dynamic World + ESA WorldCover	Land-cover support	Rapid 10 m class probabilities and fixed global land-cover maps help prototype, stratify and validate—not calculate credits directly.
NASADEM + SoilGrids + CHIRPS + ERA5-Land	Environmental controls	Terrain, soil, rainfall and climate variables reduce bias in reference matching and explain natural growth.
Project and field inventory	Ground truth and audit evidence	Planting dates, species, diameter, height, survival and pre-project status are required for defensible attribution and carbon modelling.

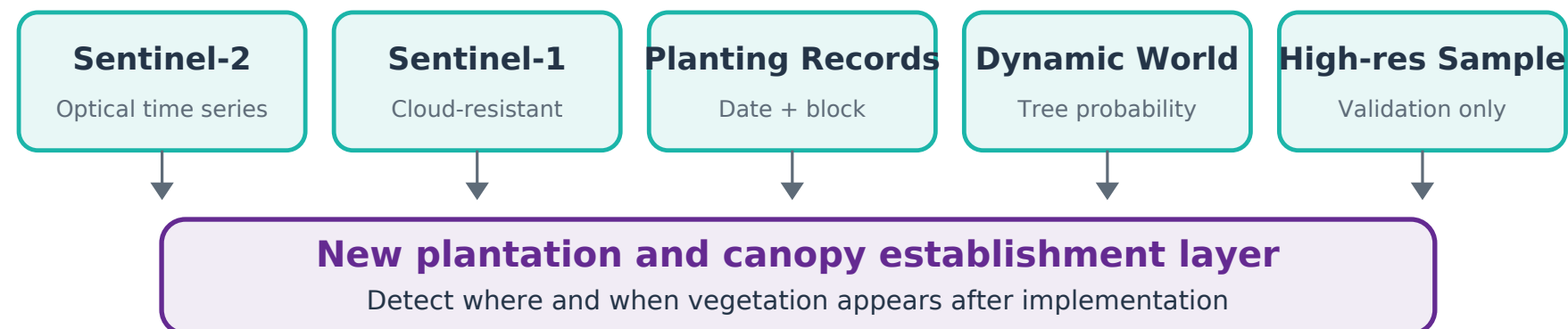
2. Dataset Combination by Project Module

A. Baseline and Pre-Existing Vegetation



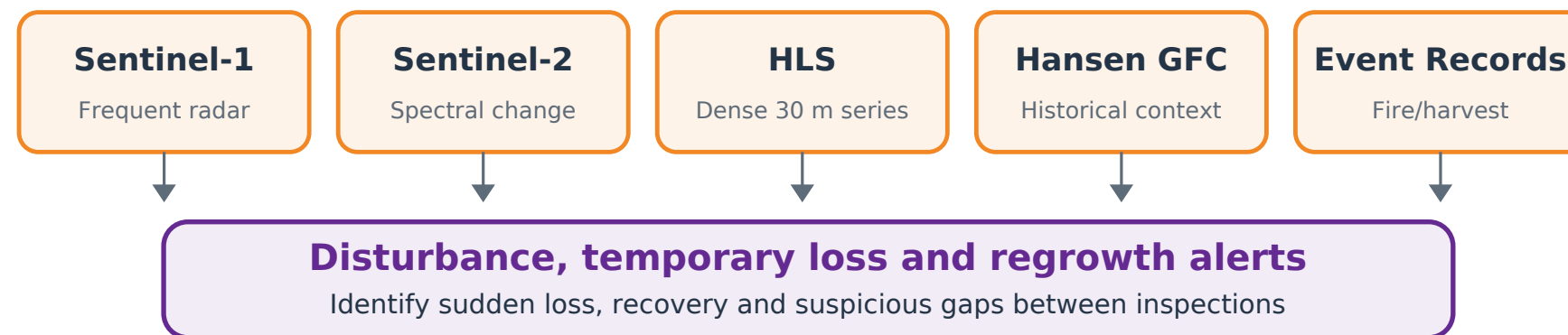
Decision: Landsat provides historical context; Sentinel-2 adds current detail; Hansen screens past loss; WorldCover supports stratification; field records identify trees that cannot be claimed as newly created carbon.

B. New Plantation Detection

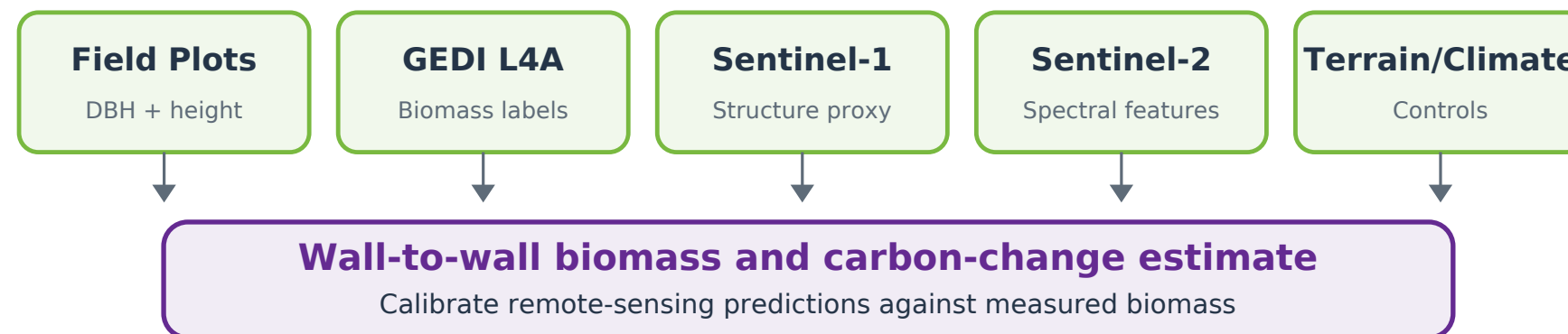


Decision: Use the project start date and planting records to anchor the time series. Treat high-resolution imagery or drones as targeted validation for uncertain zones.

C. Cutting and Regrowth Detection

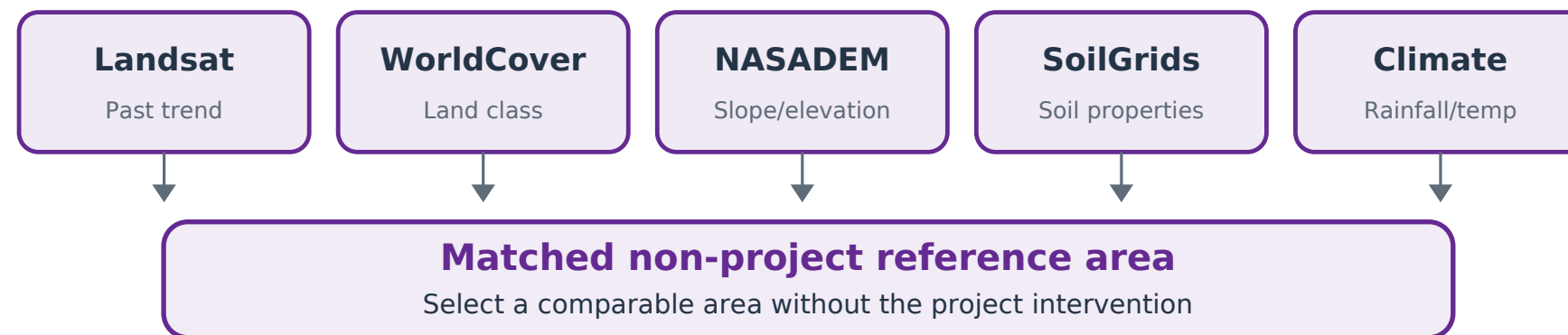


D. Biomass and Carbon Estimation

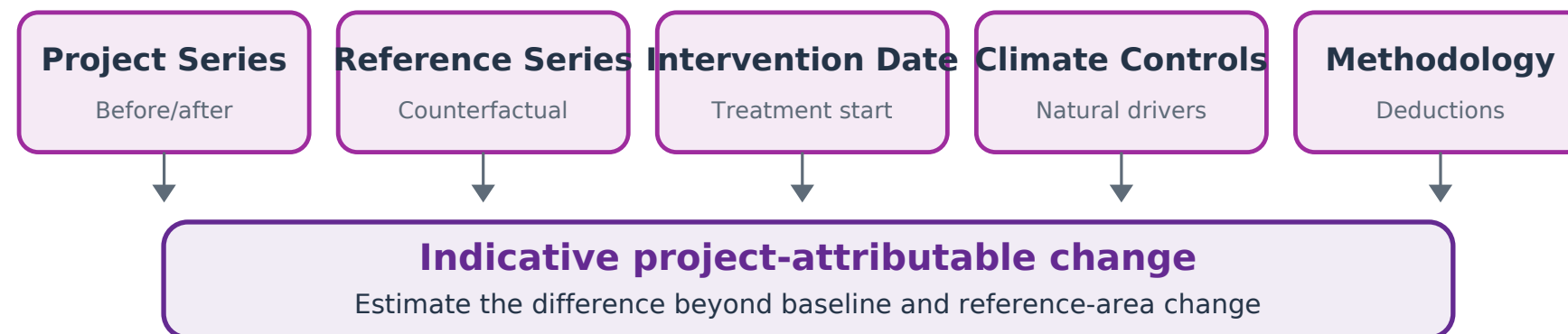


Important: Canopy cover is not biomass, biomass is not automatically eligible carbon, and eligible carbon is not automatically a verified credit. Methodology rules and expert review remain mandatory.

E. Reference-Area Matching



F. Additionality Assessment



Potentially attributable carbon
= Post-project project-area carbon
- Pre-project project-area carbon
- Reference-area change
- Methodology deductions
- Uncertainty adjustment

3. Implementation Matrix

Module	Minimum free inputs	Output	Main caution
Baseline	Landsat, Sentinel-2, Hansen GFC, WorldCover, field inventory	Pre-project canopy and carbon baseline	A recent green pixel does not reveal whether a tree predates the project.
Plantation detection	Sentinel-2, Sentinel-1, planting records, Dynamic World	New vegetation probability and establishment date	10 m pixels cannot reliably separate individual mixed trees.
Cutting/regrowth	Sentinel-1, Sentinel-2, HLS, event records	Loss, disturbance and recovery alerts	Seasonality, rainfall and radar moisture effects can create false alarms.
Biomass	Field plots, GEDI, Sentinel-1/2, terrain, climate	Biomass density and carbon-stock estimates	GEDI is sampled, not wall-to-wall; local calibration is required.
Reference matching	Historical imagery, land cover, terrain, soil, climate	Comparable non-project area and similarity score	Biased control selection can invalidate additionality.
Additionality	Project/reference series, intervention date, rules	Indicative attributable carbon with uncertainty	Change detection does not by itself prove causation.

Free open data for continuous screening -> high-resolution imagery for uncertain zones -> field inspection for confirmation -> human approval for carbon claims.

4. Starter Data Pipeline Pattern

A production implementation can use Google Earth Engine or equivalent cloud geospatial services for large-scale imagery processing. The following code pattern shows the required sequence, not a complete carbon-credit calculation.

```
# 1. Define project and reference polygons
project_aoi = load_boundary("project.geojson")
reference_aoi = load_boundary("reference.geojson")

# 2. Build aligned time-series features
s2 = load_sentinel2(project_aoi, cloud_mask=True)
s1 = load_sentinel1(project_aoi, polarization=["VV", "VH"])
landsat = load_landsat_history(project_aoi)

# 3. Establish baseline and detect change
baseline = build_preproject_baseline(landsat, s2, project_start_date)
change = detect_canopy_change(baseline, s2, s1)

# 4. Calibrate biomass using field plots and GEDI
biomass_model = train_biomass_model(field_plots, gedi, s1, s2, terrain)
project_carbon = biomass_model.predict(project_aoi)
reference_carbon = biomass_model.predict(reference_aoi)

# 5. Estimate additionality with uncertainty
result = estimate_additionality(
    project_carbon, reference_carbon, methodology_rules
)
return generate_audit_report(result, evidence_layers=True)
```

Interview explanation: Geospatial AI detects what changed, statistical comparison estimates what may be attributable to the intervention, Generative AI explains the evidence, and qualified experts approve the conclusion.

5. Constraints and Expected Outcomes

Constraint	Design response
Coarse resolution and mixed pixels	Analyse at canopy/block level; use targeted high-resolution or drone verification where individual-tree separation is essential.
Cloud cover and monsoon gaps	Fuse Sentinel-1 radar with cloud-masked Sentinel-2 composites.
Seasonality and natural growth	Compare same-season observations and include rainfall, temperature and reference-area trends.
Sparse biomass labels	Combine GEDI footprints with field plots; report uncertainty and regional transfer limitations.
Weak reference area	Use measurable similarity criteria and preserve an auditable selection score.
Certification risk	Treat outputs as decision support until methodology compliance, field validation and verifier approval are complete.

Expected technical outcome	Expected business outcome
Versioned baseline, canopy-change maps, disturbance alerts, biomass model, uncertainty intervals and evidence-linked reports.	Faster monitoring, targeted field visits, reduced over-crediting risk, stronger audit evidence and greater buyer confidence.

6. Authoritative Dataset References

Dataset specifications should be rechecked at implementation time because product versions, access policies and temporal coverage can change.

Source	Used in this guide
Sentinel-2 Earth Engine Data Catalog	Visible and NIR bands at 10 m; red-edge and SWIR at 20 m; global 5-day revisit.
Sentinel-1 GRD Earth Engine Data Catalog	C-band SAR GRD scenes with product resolutions of 10, 25 or 40 m depending on mode.
USGS Landsat Collection 2 Surface Reflectance	Delivered surface-reflectance products have 30 m spatial resolution.
USGS Landsat Collection 2	Level-2 surface reflectance products available from 1982 to present; Level-1 archive extends to 1972.
NASA/University of Maryland GEDI mission	Lidar observations of forest canopy height, vertical structure, elevation and biomass.
ESA WorldCover / Earth Engine catalog	Global 10 m land-cover maps for 2020 and 2021 with 11 classes.
NASA HLS / Earth Engine catalog	Harmonized Landsat-Sentinel observations every 2-3 days at 30 m for dense temporal analysis.

Prepared as a professional interview-revision guide. Carbon-credit eligibility and verification remain methodology- and jurisdiction-dependent.